ECONOSCRIPTS · EDITION 7

Economic Geography of Southern Africa

Author	Date	Keywords
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This edition demonstrates how spatial mapping might be a valuable approach when analysing trends across time and units. The demo proceeds in two steps.

1. In the first step, we construct a time series data visualisation using data from countries in Southern Africa.
2. Later, we show spatial mapping of the same data from the lens of decades to still show useful insights as well.

In [54]:
```
# ── EconoScripts Setup ──────────────────────────────────
source("C:/Users/lucio/OneDrive/Desktop/Professional/Newsletter/newsletter_viz/econoscripts_theme.R")

options(repr.plot.width = 10, repr.plot.height = 6, repr.plot.res = 150)
```

In [55]:
```
# Load packages
pacman::p_load("sf", "rnaturalearth", "rnaturalearthdata", "tidyverse", "WDI", "ggthemes", "scales", "ggtext")
```

In [56]:
```
#Retrieve data for the world map
world <- ne_countries(scale = "medium",
                      type = "countries",
                      returnclass = "sf")
```

In [57]:
```
# Data for Southern Africa
southern_africa <- c("AGO", "BWA", "SWZ", "LSO", "MWI", "MOZ", "NAM", "ZAF", "ZMB", "ZWE")
```

In [58]:
```
# Map for southern Africa
southern_africa_map <- world %>% filter(iso_a3 %in% southern_africa)
```

In [59]:
```
#Basic plot
options(repr.plot.width = 10, repr.plot.height = 6)
ggplot(data = southern_africa_map) +
  geom_sf() + theme_void()
```

In [60]:
```
# GDP per capita data, PPP (constant 2021 international dollars)
gdp_data <- WDI(
  country   = southern_africa,
  indicator = "NY.GDP.PCAP.PP.KD",
```

```
  start     = 1980,
  end       = 2020,
  extra     = TRUE   # pulls iso3c, region, etc.
)
```

In [61]:
```
dim(gdp_data)
```

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In [62]:
```
# Keep variables of interest only
df_clean <- gdp_data %>%
    mutate(gdp_pc = as.numeric(NY.GDP.PCAP.PP.KD)) %>%
    select(iso3c, country, year, gdp_pc) %>%
arrange(year)
```

In [63]:
```
# Check missing values
which(is.na(df_clean$gdp_pc))
```

1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10 · 11 · 12 · 13 · 14 · 15 · 16 · 17 · 18 · 19 · 20 · 21 · 22 · 23 · 24 · 25 · 26 · 27 · 28 · 29 · 30 · 31 · 32 · 33 · 34 · 35 · 36 · 37 · 38 · 39 · 40 · 41 · 42 · 43 · 44 · 45 · 46 · 47 · 48 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 56 · 57 · 58 · 59 · 60 · 61 · 62 · 63 · 64 · 65 · 66 · 67 · 68 · 69 · 70 · 71 · 72 · 73 · 74 · 75 · 76 · 77 · 78 · 79 · 80 · 81 · 82 · 83 · 84 · 85 · 86 · 87 · 88 · 89 · 90 · 91 · 92 · 93 · 94 · 95 · 96 · 97 · 98 · 99 · 100

In [80]:
```
# Print the first 20 rows
head(df_clean, n = 10)
```

A data.frame: 10 × 4

	iso3c	country	year	gdp_pc
	<chr>	<chr>	<int>	<dbl>
1	AGO	Angola	1990	8368.7229
2	BWA	Botswana	1990	10520.9184
3	SWZ	Eswatini	1990	5447.7390
4	LSO	Lesotho	1990	1422.1506
5	MWI	Malawi	1990	1047.4644
6	MOZ	Mozambique	1990	547.6228
7	NAM	Namibia	1990	7281.7789
8	ZAF	South Africa	1990	10811.1515
9	ZMB	Zambia	1990	2425.1113
10	ZWE	Zimbabwe	1990	6082.8425

In [81]:
```
# Print the last 20 rows
tail(df_clean, n= 10)
```

A data.frame: 10 × 4

	iso3c	country	year	gdp_pc
	<chr>	<chr>	<int>	<dbl>
301	AGO	Angola	2020	8960.338
302	BWA	Botswana	2020	16289.391
303	SWZ	Eswatini	2020	9695.825
304	LSO	Lesotho	2020	2517.110
305	MWI	Malawi	2020	1656.440
306	MOZ	Mozambique	2020	1466.125
307	NAM	Namibia	2020	9652.248
308	ZAF	South Africa	2020	13250.567
309	ZMB	Zambia	2020	3391.595
310	ZWE	Zimbabwe	2020	4527.720

In [66]:

```
# Removw miaing values
df_clean <- df_clean %>%
            filter(year %in% 1990:2020)
```

In [67]:

```
# Descriptive stats
df_clean %>%
  group_by(country) %>%
  summarise(
    Mean   = mean(gdp_pc, na.rm = TRUE),
    SD     = sd(gdp_pc,   na.rm = TRUE),
    Min    = min(gdp_pc,  na.rm = TRUE),
    Max    = max(gdp_pc,  na.rm = TRUE),
    N      = n()
  ) %>%
  arrange(desc(Mean)) %>%
  mutate(across(c(Mean, SD, Min, Max), ~ scales::dollar(., prefix = "$", big.mark = ","))) %>%
  knitr::kable(caption = "GDP per capita Summary Statistics by Country (1990–2020)")
```

```
Table: GDP per capita Summary Statistics by Country (1990–2020)

|country      |Mean       |SD        |Min        |Max        |  N|
|:------------|:----------|:---------|:----------|:----------|--:|
|Botswana     |$14,156.93 |$2,269.29 |$10,520.92 |$18,106.01 | 31|
|South Africa |$12,580.47 |$1,783.44 |$9,976.97  |$14,681.10 | 31|
|Namibia      |$9,360.13  |$1,684.04 |$7,281.78  |$12,166.57 | 31|
|Angola       |$8,835.12  |$2,033.76 |$5,379.48  |$11,721.66 | 31|
|Eswatini     |$7,339.19  |$1,600.80 |$5,291.90  |$10,099.59 | 31|
|Zimbabwe     |$5,150.29  |$1,015.20 |$2,954.10  |$6,582.35  | 31|
|Zambia       |$2,796.74  |$583.79   |$2,110.58  |$3,646.96  | 31|
|Lesotho      |$2,276.89  |$516.53   |$1,422.15  |$3,057.26  | 31|
|Malawi       |$1,330.91  |$219.64   |$975.85    |$1,687.29  | 31|
|Mozambique   |$1,005.71  |$376.45   |$510.82    |$1,538.84  | 31|
```

In [68]:

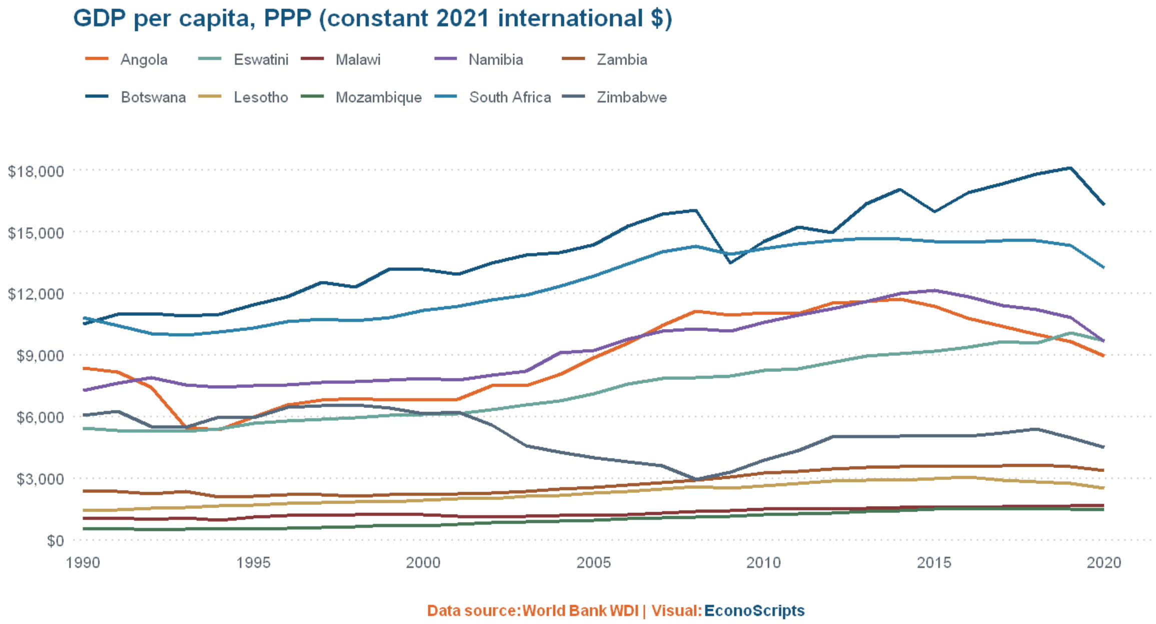
```
# Preparation for data visualisation
options(repr.plot.width = 10, repr.plot.height = 5.5)
dir.create("figures/", showWarnings = FALSE)
```

In [69]:

```
# ── Dynamic colour assignment ────────────────────────────────
n_countries <- n_distinct(df_clean$country)

es_base_colours <- c(
  "#E1692D",   # orange        — primary (Malawi anchor)
  "#14557D",   # teal dark
  "#6CA89C",   # muted teal
  "#C4A35A",   # warm gold
  "#8B3A3A",   # deep red
  "#4A7C59",   # forest green
  "#7B5EA7",   # muted purple
  "#2E86AB",   # sky blue
  "#A05C34",   # burnt sienna
  "#556B7D",   # steel grey
  "#B5563E",   # terra cotta
```

```r
  "#2D6B4F"    # deep green
)

# Recycles gracefully if n_countries > length(es_base_colours)
country_colours <- setNames(
  rep(es_base_colours, length.out = n_countries),
  sort(unique(df_clean$country))
)
```

In [70]:
```r
g1 <- df_clean %>%
  ggplot(aes(x = year, y = gdp_pc, colour = country, group = country)) +

  geom_hline(
    yintercept = seq(0, 18000, by = 3000),
    colour     = "#D3D3D3",
    linetype   = "dotted",
    linewidth  = 0.5
  ) +

  geom_line(linewidth = 1.1) +

  scale_colour_manual(values = country_colours) +

  scale_x_continuous(
    limits = c(1990, 2020),
    breaks = seq(1990, 2020, by = 5),
    expand = expansion(mult = c(0.01, 0.05))  # reduce right padding — no inline labels
  ) +

  scale_y_continuous(
    breaks = seq(0, 18000, by = 3000),
    labels = label_dollar(prefix = "$", big.mark = ","),
    expand = expansion(mult = c(0.02, 0.08))
  ) +

  labs(
    title   = "GDP per capita, PPP (constant 2021 international $)",
    y       = NULL,
    colour  = NULL,
    caption = es_caption("World Bank WDI | Visual: ")
  ) +

  theme_econoscripts() +
  theme(
    axis.line.y       = element_blank(),
    legend.position   = "top",                 # clean top legend
    legend.justification = "left",         # ← align left
    legend.direction = "horizontal",
    legend.text       = element_text(size = 9, colour = es$grey_text),
    legend.key        = element_blank(),
    legend.spacing.x = unit(0.4, "cm"),

    # Caption centred with more space above
    plot.caption = element_markdown(
      colour   = es$orange,
      face     = "bold",
      size     = 9,
      hjust    = 0.5,
      margin   = margin(t = 20, b = 4)
    )
  )
g1
```

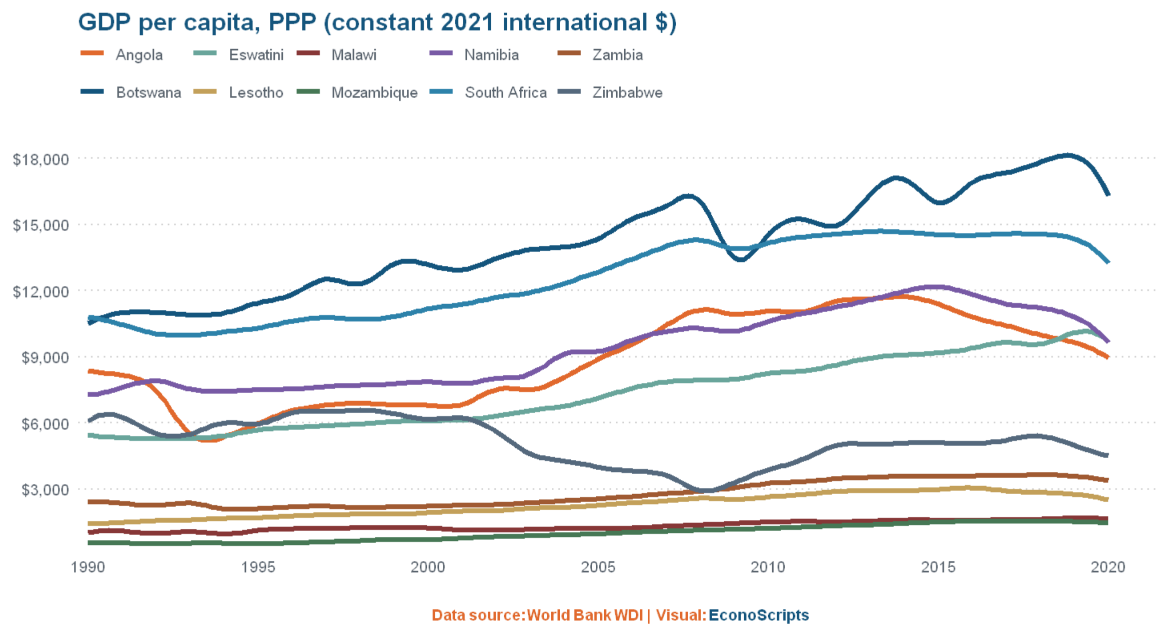


In [71]:
```
# —— Export ——————————————————————————————————————————————————————————
ggsave(
  filename = "figures/Edition_8a.png",
  plot     = last_plot(),
  width    = 10,
  height   = 5.5,
  units    = "in",
  dpi      = 300,
  bg       = "white"
)
```

In [72]:
```
# Smooth the data first
df_smooth <- df_clean %>%
  group_by(country) %>%
  reframe({
    sp <- spline(year, gdp_pc, n = 200)
    data.frame(year = sp$x, gdp_pc = sp$y)
  })
```

In [73]:
```
# Plot using smoothed data

g2 <- df_smooth %>%
  ggplot(aes(x = year, y = gdp_pc, colour = country, group = country)) +

  geom_line(linewidth = 1.5) +

  scale_colour_manual(values = country_colours) +

  scale_x_continuous(
    limits = c(1990, 2020),
    breaks = seq(1990, 2020, by = 5),
    expand = expansion(mult = c(0.01, 0.05))
  ) +

  scale_y_continuous(
    breaks = seq(0, 18000, by = 3000),
    labels = label_dollar(prefix = "$", big.mark = ","),
    expand = expansion(mult = c(0.02, 0.08))
  ) +

  labs(
    title   = "GDP per capita, PPP (constant 2021 international $)",
    y       = NULL,
    colour  = NULL,
    caption = es_caption("World Bank WDI | Visual: ")
  ) +

  theme_econoscripts() +
  theme(
    legend.position      = "top",
    legend.justification = "left",
```

```
    legend.direction     = "horizontal",
    legend.text          = element_text(size = 9, colour = es$grey_text),
    legend.key           = element_blank(),
    legend.spacing.x     = unit(0.4, "cm"),
    legend.margin        = margin(l = 0),

    # Caption centred with more space above
    plot.caption = element_markdown(
      colour  = es$orange,
      face    = "bold",
      size    = 9,
      hjust   = 0.5,
      margin  = margin(t = 20, b = 4)
    )
  )
g2
```



In [74]:
```
# —— Export ——————————————————————————————————————————————————————————————————
ggsave(
  filename = "figures/Edition_8b.png",
  plot     = last_plot(),
  width    = 10,
  height   = 5.5,
  units    = "in",
  dpi      = 300,
  bg       = "white"
)
```

In [75]:
```
# Filter to your decade snapshots
gdp_decades <- df_clean %>%
  filter(year %in% c(1990, 2000, 2010, 2020)) %>%
  select(iso3c, country, year, gdp_pc) %>%
  arrange(year)
```

In [76]:
```
# Joining data
southern_africa_data <- southern_africa_map %>%
  left_join(gdp_decades, by = c("sov_a3" = "iso3c"))
```

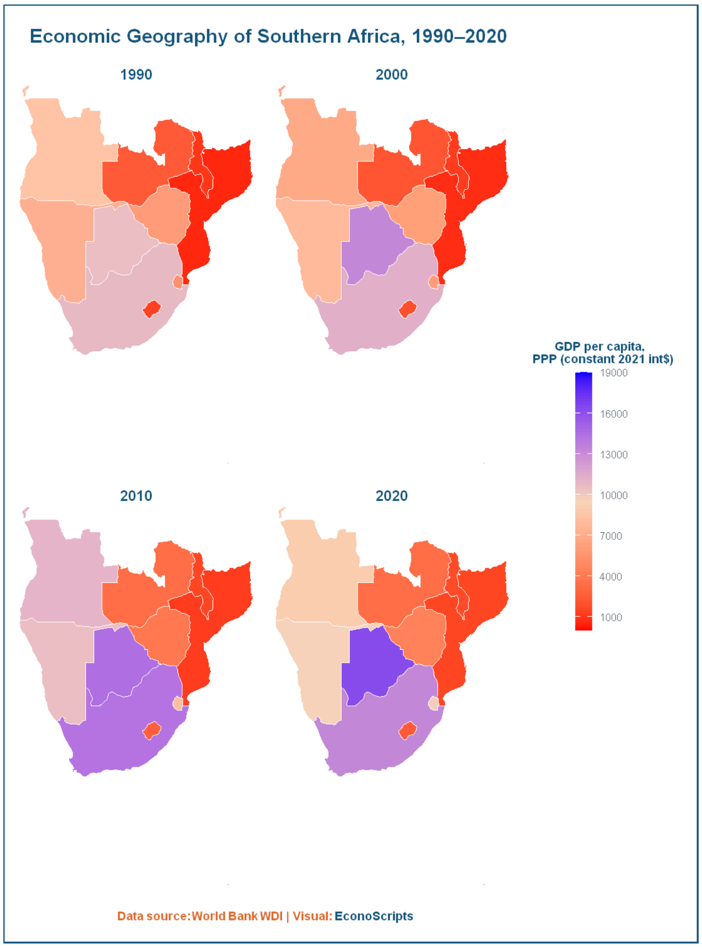
In [77]:
```
# Check for failed joins — these will appear as NA in gdp_pc
southern_africa_data %>%
  filter(is.na(gdp_pc)) %>%
  select(sov_a3, year) %>%
  print()
```

```
Simple feature collection with 0 features and 2 fields
Geometry type: MULTIPOLYGON
Bounding box:  xmin: NA ymin: NA xmax: NA ymax: NA
Geodetic CRS:  WGS 84
[1] sov_a3    year      geometry
<0 rows> (or 0-length row.names)
```

In [78]:

```r
# Mapping

options(repr.plot.width = 14, repr.plot.height = 10, repr.plot.res = 150)

g3 <- ggplot(southern_africa_data) +
  geom_sf(aes(fill = gdp_pc), color = "white", linewidth = 0.3) +

  scale_fill_gradient2(
    low      = "red",
    mid      = "#FAD5BC",
    high     = "blue",
    midpoint = 9500,
    name     = "GDP per capita, \n PPP (constant 2021 int$)",
    na.value = "grey88",
    breaks   = c(1000, 4000, 7000, 10000, 13000, 16000, 19000),  # ← updated
    limits   = c(0, 19000),                                       # ← updated
    guide    = guide_colorbar(
      barheight   = unit(7, "cm"),
      barwidth    = unit(0.45, "cm"),
      title.hjust = 0.5,
      title.theme = element_text(
        size   = 9,
        face   = "bold",
        colour = es$teal,
        hjust  = 0.5,
        margin = margin(b = 6)
      ),
      label.theme = element_text(size = 8, colour = es$grey_text)
    )
  ) +

  facet_wrap(~ year, ncol = 2) +

  coord_sf(expand = FALSE) +

  labs(
    title   = "Economic Geography of Southern Africa, 1990–2020",
    caption = es_caption("World Bank WDI | Visual: ")
  ) +

  theme_void(base_size = 12) +
  theme(
    plot.background = element_rect(
      fill      = "#FFFFFF",
      colour    = es$teal,
      linewidth = 1.0
    ),

    plot.title = element_text(
      face   = "bold",
      size   = 15,
      colour = es$teal,
      hjust  = 0,
      margin = margin(t = 6, b = 14, l = 8)
    ),

    plot.caption = element_markdown(
      size   = 9,
      colour = es$orange,          # ← orange per updated theme
      face   = "bold",             # ← bold per updated theme
      hjust  = 0.5,                # ← centred
      margin = margin(t = 20, b = 6)
    ),

    strip.text = element_text(
      face   = "bold",
      size   = 11,
      colour = es$teal,
      margin = margin(b = 4, t = 2)
    ),

    panel.background = element_blank(),

    legend.position      = "right",
    legend.justification = "center",
    legend.text          = element_text(size = 8, colour = es$grey_text),
```

```
    legend.title         = element_text(
      size   = 9,
      face   = "bold",
      colour = es$teal,
      hjust  = 0.5
    ),
    legend.margin   = margin(l = 12, r = 8),
    legend.box.just = "center",

    panel.spacing = unit(0.6, "cm"),
    plot.margin   = margin(12, 12, 10, 12)
  )
g3
```



In [79]:
```
# — Export ———————————————————————————————————————————————————————————
ggsave(
  filename = "figures/Edition_8c.png",
  plot     = last_plot(),
  width  = 14,
  height = 10,
  units    = "in",
  dpi      = 300,
  bg       = "white"
)
```

Best regards,

Lucious